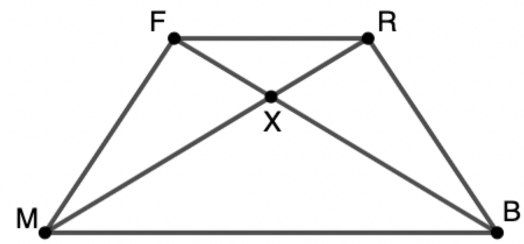
Geometry | Name____________________
Unit 8 | Date____________________
Lesson 5 Practice | Period____________________

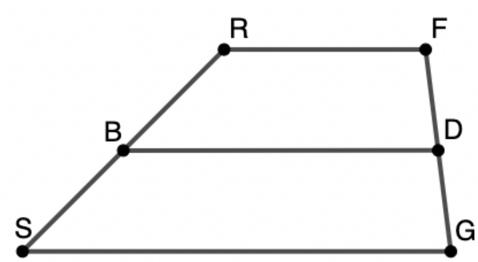
Given: $FRBM$ is an isosceles trapezoid, $\angle MFX \cong \angle BRX$.

Prove: ΔMXB is an isosceles triangle.



Statement	Reason
$FRBM$ is an isosceles trapezoid	Given
$\angle FMB \cong \angle RBM$	1.
$\angle MFX \cong \angle BRX$	Given
2.	Reflexive Property of Congruence
3.	AAS Congruence
4.	Corresponding Parts of Congruent Triangles are Congruent
ΔMBX is an isosceles triangle	5.

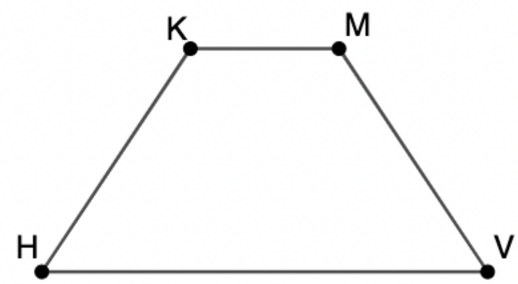
Trapezoid $SRFG$ is shown. $\overline{BD}$ is a midsegment of $SRFG$.
$FD = 4x + 34$, $DG = 8x - 14$, $RB = 8y + 41$, $SB = 13y - 4$.



6. Find the values of x and y.

7. Find the lengths of $\overline{RS}$ and $\overline{DG}$.

Isosceles trapezoid $KMVH$ is shown where $m\angle KHV = (16z - 6)°$, $m\angle HKM = (21z + 38)°$, $m\angle KMV = (16t - 26)°$.



8. What is the value of z?

9. What is the value of t?

10. What are $m\angle VHK$, $m\angle HKM$, $m\angle KMV$, and $m\angle MVH$?